Interpretation of Structural Circularity in the TNA Framework

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A Structural Mapping from Formal Logic to Dynamic Systems

Abstract

We interpret the structural circularity identified in formal theories as a fundamental property of generative systems. Within the TNA framework, circularity is not a defect but a necessary condition for persistence and expansion. We map formal non-self-sufficiency into boundary-driven dynamics, where indeterminacy generates expansion pressure. This establishes a bridge between formal logic and dynamic epistemic systems.

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1. Starting Point

From the formal result:

> *A theory cannot validate its own rules without either circularity or external semantics.*

We reinterpret this as a **structural constraint**:

- systems cannot fully ground themselves internally
- closure requires either:
    - external input, or
    - recursive structure

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2. Mapping into TNA

We define a correspondence between formal logic and TNA:

```
Formal Logic          →   TNA

Theory T              →   System Σ
Inference Rules R     →   Structural Operators
Semantics M           →   External Structure (N₀)
Derivability          →   Internal Consistency (R)
Circularity           →   Structural Recursion
```

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3. Core Interpretation

Definition (TNA System)

A system is:

```
Σ = (Ω, ∂Ω, R, H_s)
```

where:

- Ω → domain of valid states
- ∂Ω → boundary of validity
- R → reliability function
- H_s → structural history

Interpretation Rule

The inability of a theory to self-validate corresponds to:

> *the existence of $\partial\Omega \neq \emptyset$*

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4. Circularity as Structural Necessity

Proposition 1

If a system attempts full internal closure, then:

```
Closure → ∂Ω → Indeterminacy
```

Interpretation

- Internal validation creates recursive dependency
- Recursive dependency generates indeterminacy
- Indeterminacy defines the boundary

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5. From Circularity to Dynamics

We define:

```
I_B = Indeterminacy at ∂Ω
F_B = Boundary pressure
```

Structural Transition

$$\text{Circularity} \;\Rightarrow\; I_B > 0 \;\Rightarrow\; F_B > 0$$

Interpretation

- Circularity is not static
- It generates **pressure for expansion**

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6. Expansion Mechanism

Rule (TNA Expansion)

```
if F_B > τ:
    Ω → Ω⁺
```

Meaning

- the system extends its domain
- new structure is introduced
- previous limits become internal

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7. External Semantics as N_0

From the formal paper:

> *rule validation requires external semantics*

In TNA:

```
External Semantics → N₀
Internal System    → N₁
```

Interpretation

- N_0 provides grounding
- N_1 provides structure
- interaction drives evolution

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8. Structural Equivalence

Theorem (Interpretative Equivalence)

The following are structurally equivalent:

$$\text{Need for external semantics} \quad \Longleftrightarrow \quad \text{Existence of boundary } \partial\Omega$$

Sketch

- If no external grounding → circularity
- Circularity → indeterminacy
- Indeterminacy → boundary

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9. Consequence: Non-Terminating Systems

Proposition 2

A system with persistent $\partial\Omega$:

- cannot reach final closure
- must continue expanding

Interpretation

$$\Sigma_t \to \Sigma_{t+1} \to \Sigma_{t+2} \to \dots$$

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10. Key Insight

The formal limitation becomes, in TNA:

> *the engine of generative systems*

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11. Minimal Structural Law

$$\boxed{\text{Circularity} \Rightarrow \text{Boundary} \Rightarrow \text{Expansion}}$$

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12. Implications

This mapping suggests:

- logic $\rightarrow$ static constraint
- TNA $\rightarrow$ dynamic realization

Applications:

- physics $\rightarrow$ singularities as $\partial\Omega$
- ML $\rightarrow$ OOD failure
- cognition $\rightarrow$ unanswerable questions

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13. Final Statement

> *What appears as a limitation in formal systems*
> *becomes a generative principle in TNA.*